

BEYOND MEAN FOILS: AUDITING WORST-FOIL SPECIFICITY IN FROZEN CLIP REGION EXPLANATIONS

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ABSTRACT

Spatial overlap and mean class contrast need not imply target-specific region responses. We audit CCI’s maximum-target-drop region in frozen CLIP, using separate prompts for selection and evaluation. Across COCO and VOC with two checkpoints, 41.16–64.78% of locality/mean-foil screen-passers fail a worst-foil criterion. With original screening memberships retained, excluding annotated-present foils leaves 39.77–63.64% failing. Exclusion reduces failure beyond exact cardinality-matched random removal by 0.86–0.92 percentage points on COCO; VOC intervals include zero. Positive-target filtering retains high failure. Among screened failures, unrestricted eight-region oracles permit sign repair in 6.25–7.84% on COCO and 27.40–31.29% on VOC. A target-drop tolerance of $\epsilon = .02$ reduces feasible repair to 0.16–0.98%; relaxing it increases opportunity with setting-dependent target and locality changes.

Index Terms— CLIP, explanation auditing, class specificity, constrained region selection

1 INTRODUCTION

CLIP [1] region explanations use intervention-induced image–text score changes. Cluster-based Concept Importance (CCI) [2] combines cluster scores into a heatmap; we audit only its maximum-target-drop region, CCI-top1. Spatial overlap, positive target contribution, and target specificity are distinct: a region may support multiple genuinely present classes.

COCOA [3] studies corpus–foil attribution; CASE [4] builds contrastive saliency maps. Related CLIP attribution [5, 6, 7], class contrasts [8], and caption foils [9] motivate our audit. Our contribution is a controlled empirical audit: we measure the mean–worst-foil gap within a declared screen, compare annotation exclusion against exact cardinality-matched removal, and decompose candidate, budget, and selector limits. Worst-foil (WF) reranking is a diagnostic probe; unlike IDEA’s few-shot adaptation [10], CLIP stays fixed.

2 AUDIT AND REPAIR OPPORTUNITY

For checkpoint m , the pre-softmax score is $s_m(x, c) = \tau_m \langle z_m(x), e_m(c) \rangle$, with unit-norm embeddings and learned

$\tau_m = \exp(\ell_m)$. Region R has intervention response

$$d_R(c) = s_m(x, c) - s_m(\mathcal{I}(x, R), c). \quad (1)$$

For target category t , selection uses “a photo of a {category}”:

$$R_{\text{CCI}} = \arg \max_{R \in \mathcal{R}(x)} d_R^{\text{sel}}(t). \quad (2)$$

Held-out margins and screens. Evaluation uses “a picture of the {category}”, “an image containing a {category}”, and “the {category}”. For unit-norm text embeddings $e_h(c)$, let $H = 3$, $n_c = \|H^{-1} \sum_h e_h(c)\|_2$, and $\bar{d}_c = H^{-1} \sum_h d_h(c)/n_c$. With $\mathcal{F}_t$ the non-target dataset categories, the aggregate worst-foil margin is

$$P(R, t) = \bar{d}_t - \max_{f \in \mathcal{F}_t} \bar{d}_f. \quad (3)$$

This equals using normalized mean text embeddings; class-specific n_c can change rankings. Tables report raw target drop $H^{-1} \sum_h d_h(t)$, whereas the positive-target screen uses $\bar{d}_t$. Evaluation templates are not used to select a region.

For CCI-top1, define $A = \{\text{bbox} \geq .5, \bar{d}_t - \text{mean}_f \bar{d}_f > 0\}$ and $F = \{P < 0\}$. The diagnostic screen and .5 threshold are study-defined, not a standard of explanation validity. Because a positive mean-foil margin does not require positive target drop, we also audit $A^+ = A \cap \{\bar{d}_t > 0\}$. The primary repair population is $B = A \cap F$, fixed before reranking. Here “failure” means failure of this declared criterion, not a semantic error or an invalid explanation.

Target-constrained diagnostic probe. Define

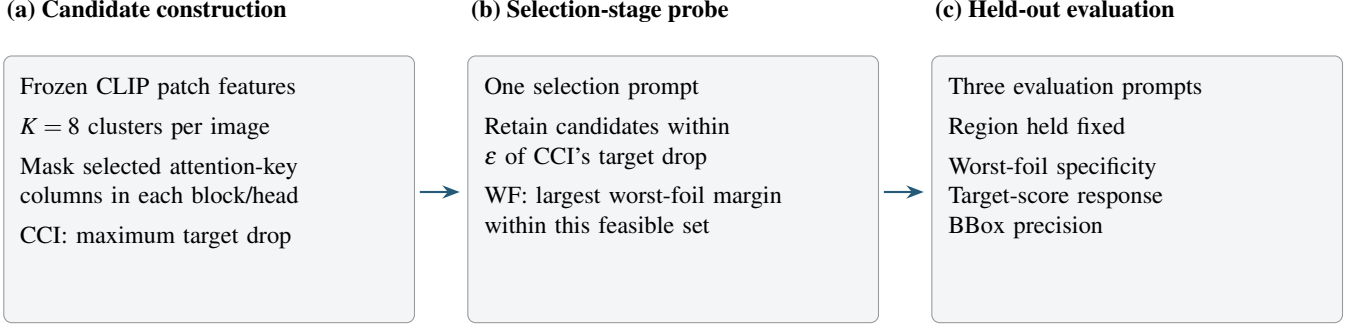
$$M^{\text{sel}}(R, t) = d_R^{\text{sel}}(t) - \max_{f \in \mathcal{F}_t} d_R^{\text{sel}}(f), \quad (4)$$

$$\mathcal{R}_\epsilon = \{R : d_R^{\text{sel}}(t) \geq d_{R_{\text{CCI}}}^{\text{sel}}(t) - \epsilon\}, \quad (5)$$

and worst-foil (WF) reranking

$$R_{\text{WF}} = \arg \max_{R \in \mathcal{R}_\epsilon} M^{\text{sel}}(R, t). \quad (6)$$

The baseline is feasible, so any switch loses at most ϵ of *selection-stage* target drop; this gives no held-out guarantee. Mean and Max-.1 instead maximize $d_t^{\text{sel}} - \text{mean}_f d_f^{\text{sel}}$



Selection uses no held-out responses; oracles inspect held-out candidates only for diagnosis.

Figure 1: Selection and held-out audit. A region is fixed using selection-stage responses before three-template evaluation of specificity, target response, and locality. The schematic depicts the protocol, not a measured response example.

and $d_t^{\text{sel}} = .1 \max_f d_f^{\text{sel}}$ over the same set. The reference budget is $\epsilon = .02$ in logit units, not 2% relative loss. A post-hoc .01/.02/.05/.10/.20 sweep selects no new optimum. Cosine budgets ϵ/τ_m are not calibrated across checkpoints; unrestricted oracles remove the budget entirely.

Exact diagnostic oracles. For a baseline-failure population B , the feasible repair count is

$$N_O(B, \epsilon) = \sum_{i \in B} \mathbf{1} \left\{ \max_{R \in \mathcal{R}_\epsilon(i)} P_i(R) \geq 0 \right\}. \quad (7)$$

The unrestricted oracle searches all K candidates. A sign pass means $P_i(R) \geq 0$, without locality or positive-target constraints. Failures split into: no passing candidate (C_0); passing candidates but *none feasible* (C_1); a feasible pass missed by WF (C_2); and WF sign repair (r). Thus $|B| = C_0 + C_1 + C_2 + r$ and $N_O = C_2 + r$. We additionally count repairs satisfying $J(R) = \{P_i(R) \geq 0, A(R), \bar{d}_t(R) > 0\}$; $A(R)$ applies the same full-foil screen to the new region, without changing original B . Oracles use held-out data only for diagnosis.

Fixed-population foil controls. For image i , let $\mathcal{F}_i^{\text{abs}}$ remove all annotated-present categories from the non-target foils. The protocol holds R_{CCI} and full-foil screen A fixed; $\mathcal{F}_i^{\text{abs}}$ tests $\bar{d}_t - \max_{f \in \mathcal{F}_i^{\text{abs}}} \bar{d}_f < 0$. No region or screen is re-selected. Reducing the foil set cannot increase failure, so we also match its cardinality. For N_i full foils, $k_i = |\mathcal{F}_i^{\text{abs}}|$, and m_i foils exceeding the target, a uniform size- k_i subset without replacement fails with probability

$$q_i = 1 - \frac{\binom{N_i - m_i}{k_i}}{\binom{N_i}{k_i}}. \quad (8)$$

With zero numerator when $k_i > N_i - m_i$, we compare $\Delta_A = |A|^{-1} \sum_{i \in A} (\mathbf{1}\{F_i^{\text{abs}}\} - q_i)$. Restricted sets are nonempty. Annotation absence does not establish semantic absence; crowd annotations remain excluded.

3 PROTOCOL AND REPRODUCIBILITY

COCO val2014 [11] and VOC2007 test [12] provide 27,708/1,943 eligible images, each evaluated with OpenAI CLIP B/16 and B/32 (59,302 image–model records). The target is the largest instance’s category, using annotation area on COCO and box area on VOC. We require another category and 2–80% target area; COCO crowd annotations are excluded. Full foils are the other 79/19 classes.

K-means clusters L2-normalized projected patch features: $K = 8$, three initializations, 50 iterations, seed 1701+ i (sorted IDs shuffled with seed 1701 define i). Intervention $\mathcal{I}$ adds $-\infty$ to attention logits at selected patch-key columns before softmax in every block/head, retaining the class-token column. Ties use first argmax; MPS features are cast to float32. BBox precision is the fraction of selected patch centers in target-class box unions, mapped by $224/\min(W, H)$ and center-crop offsets; it measures neither pixel precision nor target coverage [13].

Float32 candidate analyses use stored responses. Foil controls regenerate full-class responses: IDs, CCI labels, feasible sizes, and full-foil states match; compared summaries differ by less than 10^{-5} . Original A, A^+ memberships are retained. This verifies summaries, not spatial identity: original masks are unavailable. Region-fixed comparisons are conditional on faithful regeneration.

Intervals use 10,000 image bootstraps and percentile 2.5/97.5 limits, pointwise without multiplicity correction. Paired quantities are differenced within image before re-sampling. Foil comparisons resample within fixed A/A^+ ; repair-cost intervals condition on actual repairs. Exact q_i is primary; 100 seeded draws check it.

4 RESULTS

Table 1: Fixed original A : full-foil, annotation-absent, and exact cardinality-matched random failure rates (%). Δ_A is absent minus random in percentage points, with paired 95% CI.

Setting	Full	Absent	Random	Δ_A
COCO B/16	64.37	63.17	64.09	-0.92 [-1.08,-0.77]
COCO B/32	64.78	63.64	64.50	-0.86 [-1.01,-0.72]
VOC B/16	42.27	40.94	41.46	-0.52 [-1.16,+0.06]
VOC B/32	41.16	39.77	40.32	-0.55 [-1.19,+0.04]

Failure persists after annotation exclusion. Screen sizes are 17,726, 17,847, 1,209, and 1,227 in table order. Exclusion resolves only 213, 203, 16, and 17 full-foil failures; 96.63–98.24% remain (failure counts in Table 2). The rate falls by 1.20/1.14/1.32/1.39 percentage points. Table 1 separates this change from foil-count reduction: COCO exclusion has a larger effect than random removal of the same number of categories, whereas VOC Δ_A intervals include zero. High failure is therefore not explained solely by explicitly annotated co-occurring classes under this protocol.

Positive-target and scene-count checks. Requiring positive target drop leaves 16,855/17,330/1,186/1,215 images. Full-foil failure in A^+ is 62.64/63.79/41.40/40.74%; annotation-absent failure remains 61.38/62.61/40.05/39.34%. Matched-random comparisons retain the COCO/VOC interval pattern. Even scenes with exactly two annotated categories have COCO annotation-absent failure of 63.39/64.95%.

Table 2: Exact local decomposition of *screened failures* $B = A \cap F$ at $\epsilon = .02$. Counts in C_0, C_1, C_2, r are mutually exclusive; $C_2 + r$ is the feasible-oracle count.

Setting	$ B $	C_0	C_1	C_2	r
COCO B/16	11,410	10,697	695	0	18
COCO B/32	11,561	10,655	875	1	30
VOC B/16	511	371	135	0	5
VOC B/32	505	347	155	1	2

Candidate and budget bottlenecks differ by dataset. No region among the eight candidates passes for 93.75/92.16% of COCO screened failures, versus 72.60/68.71% on VOC (Table 2). Removing the budget would therefore permit sign repair in only 6.25/7.84% of COCO B , versus 27.40/31.29% on VOC. Figure 2 summarizes capacity within these eight candidates. The .02 budget then excludes 96.43–98.10% of those unrestricted opportunities: for COCO B/16, 695 of 713 are excluded, leaving 18. Across settings, feasible-oracle counts are 18/31/5/3 (0.16/0.27/0.98/0.59% of B), and WF finds 18/30/5/2.

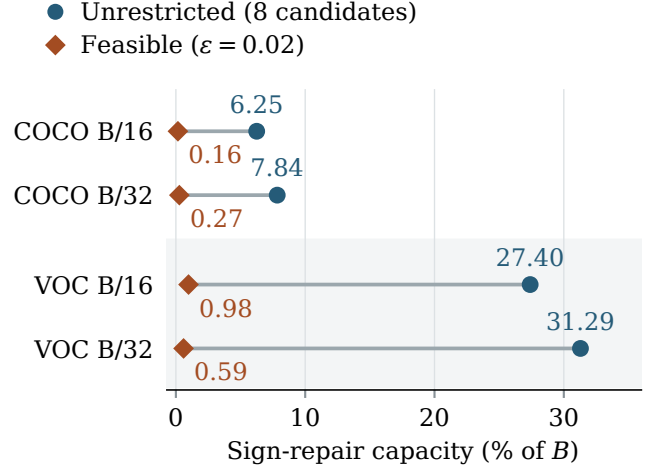


Figure 2: Budget-constrained sign-repair capacity among fixed screened failures B . Connected markers compare the same setting on a shared linear scale. Unrestricted: at least one of all eight candidates has $P \geq 0$; feasible: at least one also meets $\epsilon = .02$. Percentages use each setting’s own $|B|$ (Table 2); these are exact counts, not margin gains or confidence intervals.

Table 3: Budget endpoints (C: COCO; V: VOC). Multi: all-image percentage with multiple feasible candidates. O_B, r_B : oracle/actual repair percentages of fixed B . $\Delta d_t, \Delta b$: all-image raw-target and bbox changes (bbox in percentage points).

Set	ϵ	Multi	O_B	r_B	Δd_t	Δb
C16	0.02	4.71	0.16	0.16	+0.0001	-0.12
C16	0.20	35.78	1.57	1.50	-0.0140	-1.63
C32	0.02	4.40	0.27	0.26	-0.0001	-0.08
C32	0.20	34.96	2.05	2.00	-0.0143	-0.60
V16	0.02	4.79	0.98	0.98	-0.0008	-0.34
V16	0.20	34.48	6.65	6.07	-0.0223	+0.08
V32	0.02	4.73	0.59	0.40	-0.0007	+0.02
V32	0.20	34.59	7.13	6.73	-0.0164	+1.11

Budget relaxation helps in every setting. At .02, 95.21–95.60% of images have only one feasible candidate (Table 3); WF has little room to change the region. At .20, oracle repair rises to 1.57/2.05/6.65/7.13% of fixed B . Raw-target changes become negative in all four settings, but bbox changes are negative on COCO and positive on VOC. Thus relaxation does not imply uniform locality loss. These post-hoc logit-budget comparisons are within-setting diagnostics, not calibrated cross-model rankings.

Global impact is not the cost of repair. Table 3 includes unchanged regions. In contrast, Table 5 conditions on actual sign repairs within B ; these conditional changes are neither global nor causal effects.

Table 4: Mean aggregate-margin differences over all eligible images, WF minus each alternative, with paired 95% CIs. Values are multiplied by 10^3 ; D counts WF/Max-.1 selection disagreements at $\epsilon = .02$.

Setting	WF-Mean	WF-Max-.1	D
COCO B/16	12.461 [10.879,14.138]	0.245 [0.128,0.374]	84
COCO B/32	9.343 [7.986,10.801]	0.241 [0.083,0.414]	89
VOC B/16	4.480 [2.318,7.054]	0.507 [0.031,1.111]	5
VOC B/32	3.184 [1.467,5.133]	0.418 [0.058,0.927]	5

Small baseline gains arise from few disagreements. WF and Max-.1 differ on only 84/89/5/5 images (Table 4). Their agreement is over 99.6% overall but 93.56/92.70/94.62/94.57% within multi-candidate sets. On the 84 COCO B/16 disagreements, WF gains .0807 margin [.0443, .1171], while raw target drop changes by $-.0326$ [$-.0628$, $-.0033$]. The whole-sample gain is .000245. The VOC disagreement subsets contain only five images each.

Table 5: Actual sign repairs *within original B*, $\epsilon = .02$. r counts repairs; j counts those also satisfying J . WF-CCI means with repair-subset 95% CIs. Δd_t is raw held-out target-score change; only bbox change is in percentage points.

Setting	r	j	Δd_t	100 Δ bbox
COCO B/16	18	12	0.029 [-0.095,0.148]	-19.53 [-38.61,-1.17]
COCO B/32	30	25	0.094 [0.037,0.149]	-14.04 [-27.61,-1.48]
VOC B/16	5	3	-0.021 [-0.170,0.182]	-16.87 [-26.11,-7.00]
VOC B/32	2	1	-0.078 [-0.115,-0.042]	-28.17 [-78.57,22.22]

Sign repair need not retain the other endpoints. Only 12/25/3/1 of the 18/30/5/2 repairs in B also retain the original screen and positive target contribution (Table 5). The joint feasible-oracle counts are 12/26/3/2. Small VOC repair subsets make their cost intervals descriptive, not evidence of stable population effects. Across *all baseline failures*, WF sign-repairs 42/48/11/6 cases; 0/4/0/0 opposite-sign regressions leave net reductions of 42/44/11/6. Across all 663/614/50/38 switches, 611/532/46/35 improve margin, with B/16 bbox changes of $-4.95/ -13.20$ points on COCO/VOC. These are different populations from B repairs. Raw-target intervals containing zero do not establish equivalence.

Table 6: Nonzero-threshold sensitivity on fixed original A . Entries are percentages with $P < -\delta$ using the normalized aggregate margin. Thresholds are post-hoc diagnostics in logit units, not calibrated semantic cutoffs.

Setting	$\delta = 0$	.01	.10	.50
COCO B/16	64.37	63.94	59.49	36.05
COCO B/32	64.78	64.38	60.22	35.19
VOC B/16	42.27	41.44	36.23	13.73
VOC B/32	41.16	40.34	34.47	11.98

Failure is not solely a near-zero effect. At $\delta = .10$, 59.49/60.22% of COCO and 36.23/34.47% of VOC screen-passers still fail (Table 6). Rates fall further at .50, so the magnitude and threshold matter. This checks sign-boundary sensitivity within the existing normalized score, not sensitivity to class-dependent normalization or semantic validity.

5 INTERPRETATION AND LIMITATIONS

Matched-cardinality controls separate annotation exclusion from foil-count reduction within a setting. Missed labels and related concepts can remain hard foils; negative margins do not establish semantic error. Hardest-foil inspection and human validation remain necessary.

The audit concerns CCI-top1, eight candidates, and one attention intervention. The COCO/VOC contrast also changes foil count (79 versus 19), target selection, and annotation conditions; it cannot isolate intrinsic dataset difficulty. Within-setting controls do not remove these differences. Other K , candidate constructions, interventions, and area-matched controls remain untested. Bbox precision measures neither coverage nor segmentation quality.

Reproducibility and normalization. Stored candidate summaries and analysis scripts support the reported threshold and budget reanalyses. Original spatial masks and complete per-class responses are unavailable in these artifacts. Consequently, summary agreement does not verify original region identity, and raw versus normalized worst-foil rankings cannot be reconstructed. Prompt holdout separates templates, not models or images. Exact candidate enumeration does not resolve these evidence gaps.

6 CONCLUSION

Under a declared worst-foil criterion, substantial failure remains after annotation-based foil exclusion and after moving the failure threshold away from zero. This is evidence about the audit criterion, not semantic error. Eight-candidate enumeration separates missing candidates from budget and selector limits. Looser budgets increase repair opportunity in every setting, with setting-dependent endpoint changes. Region identity, normalization, and foil-ontology controls are needed before broader claims about explanation reliability or cross-dataset differences.

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COMPLIANCE WITH ETHICAL STANDARDS

This work analyses publicly released COCO and PASCAL VOC images and annotations. No participants were recruited and no new human or animal experiments were conducted.

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